INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

MILL, VERTICAL AUTOMATED MANUFACTURING MACHINE, MACHINING CENTER SINGLE COLUMN, SINGLE-SPINDLE, COMPUTER NUMERICAL CONTROL (CNC)

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) covers computer numerical control (CNC) multi-axis, single-spindle, single-column, vertical machining centers intended for accurately and repeatedly shaping workpieces (i.e., raw blocks of materials) by the subtractive manufacturing material machining processes. CNC is in reference to a computer-controlled system able to programmatically command and control servomotors, facilitating various movements and functions of machine axes. Machining process examples described herein are drilling, reaming, tapping, threading, counterboring, countersinking, pocketing, squaring, stepping, plain milling, angular milling, face milling, contouring, profiling, gear hobbling and boring.

The equipment identified herein shall be manufactured in the United States of America.

CNC milling is an automated machine process used to manufacture engineering parts by selectively and progressively removing material from a workpiece. Material removal (milling) occurs via use of a rotating multi-point cutting tool and computerized controls (i.e., CNC). Parts manufactured via CNC milling are most practical via use of Computer-Aided Design (CAD) and Computer-Aided Manufacturing (CAM) software. CAD enables development and design of a part. CAM programs are used to generate manufacturing tool paths and to automate the manufacturing processes; CAM converts the CAD model into an executable CNC program (.NC) for machine use. This results in CNC manufacturing (machine control), instructing (via "G-code") the machine to execute the job in order to manufacture a physical part.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: AFLCMC/EZSS, 2145 Monahan Way, Area B, Bldg 28, Wright-Patterson AFB, OH 45433 or email engineering.standards@us.af.mil. Since contact information can change, you should verify the currency of this information using the ASSIST Online database at https://assist.dla.mil.

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G-code, per Electronic Industries Association (EIA) SAE RS-274, is required as the programming language; it describes geometric paths using a simple command structure of the letter “G” followed by a numeric value.

Multi-axis, as identified herein, represents three through five axis as per the following:

3 axis: The workpiece remains in the same position while the cutting tool operates along the X (left to right movement), Y (front to back movement) Z (up and down movement) plane to trim away material.

4 axis: The workpiece is machined in the same manner as with a three axis machine, but has an added 180 degree rotational movement capability around a single axis (example: Y axes). This is considered the fourth Axis. The axis can be either set to continually move/rotate and/or incrementally rotate/hold/lock into position during milling. The addition of a fourth axis allows a workpiece to be flipped over (180 degrees), so the machine can remove material from the bottom side.

5 Axis: The workpiece can be automatically be manipulated/adjusted from five sides at one time. The machine operates similar to a four axis mill but with an additional rotational axis so that two axis have the capability to rotate the piece up to 180 degrees at once. Application supports complex components (ex: cast parts). Requires greater CNC programming preparation time to facilitate complex rotatory movement, but allows one workpiece to be worked on all five sides during one operation.

2. CLASSIFICATION. The machining centers shall be of the following types (see 7.3).

Type I – 3 Axis, 6,000 Spindle revolutions per minute (rpm), 7.5 horsepower (hp), 10 tool capacity (see Table I).

Type II – 3 Axis, 4,000 Spindle rpm, 1 tool capacity, 7.5 hp, 1 tool capacity (see Table I).

Type III – 3 Axis, 8,000 Spindle rpm, 30 hp, 20 tool capacity (see Table I).

Type IV – 5Axis, 20,000 Spindle rpm, 24 hp, 30 tool capacity (see Table I).

Type V – 5 Axis, 8,100 Spindle rpm, 30 hp, 30-50 tool capacity (see Table I).

The milling machines shall comply with the applicable minimum requirements in Table I.

3. SALIENT CHARACTERISTICS.

3.1 General requirements. The machining center shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, or repair. The machining center and CNC system shall be compatible in design and integrated into a complete functional unit. The CNC system shall direct all tool changes, spindle operations, and linear rotational motion on all axes for continuous-path, full-contour machining. The term "full contouring" is defined herein as a closed-loop control system having simultaneous control of position, direction, and rate of movement by linear, parabolic, circular, or helical interpolation. The machining center and CNC system shall be shipped to the AF ready to operate pending training and installation. The term “ready to operate” is defined herein as a minimum set of

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tooling and accessories, internal and external hardware and software, and components required for installation to operate the machining center and CNC systems to perform milling operations. The machining centers and CNC systems identified herein shall be fully enclosed and be designed to operate unattended at all times. The contractor shall identify all risks associated with unattended operations, and identify risk mitigation procedures and recommendations in the event of malfunction to ensure and maximize/maintain human and equipment safety.

3.2 Components. The machining center shall include, as a minimum, the following components:

3.2.1 Base. The base shall support the other elements of the machine and shall incorporate materials, shape, and mass to minimize adverse vibration and deflection. The base shall provide a means for leveling and anchoring the machine.

3.2.2 Worktable. The worktable length, width, and workpiece shall be in accordance with Table I. The number and size of worktable T-slots shall be as specified in Table I. The configuration of the T-slots shall conform to the requirements of American National Standards Institute (ANSI) B5.1.

3.2.3 Spindle assembly. The machining center shall be equipped with the manufactures standard spindle head, spindle speed range, and spindle drive motor. The spindle taper and tool shank V-flange shall comply with ANSI B5.50. The spindle head shall be cooled (e.g., air, water, oil, or refrigerant cooling) to control spindle thermal growth.

3.2.4 Machineways. Machineways for the worktable, cross-slide, column, and spindle head shall be of the design considered by the manufacturer as standard for the machine. All machineways shall have telescoping protective covers to keep chips, coolant, and other debris from contacting bearing surfaces.

3.2.5 Enclosure panels and splash guards. The machining center shall be equipped with enclosure panels and splash guards.

3.2.6 Automatic tool changer and storage magazine. Where applicable per Table I, the machining center shall have an automatic tool changer and tool storage magazine. The automatic tool changer shall be capable of selecting the next tool from the storage magazine using the bidirectional random shortest path method. The automatic tool changer shall be capable of exchanging tools, returning the previous tool to the storage magazine, and tracking the tool for reuse.

3.2.7 Rigid tapping. Rigid tapping shall be provided.

3.2.8 Pressurized air system. The machining center shall provide an air supply piping system that can be connected to a pressurized air system. The piping system shall direct pressurized air through the spindle and to tool adapter tapers to remove debris from the taper surfaces during the tool-changing process.

3.2.9 Chip conveyor system. Workpiece chips shall be removed automatically from the work area, and a conveyor system shall be provided to transfer chips to a collection bin.

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3.2.10 CNC system. The CNC system shall be based on either a personal computer (PC) open architecture or other software-programmable microprocessor design. System hardware and control panels shall be industrial grade.

3.2.11 Milling machine features. A convenience package shall be included and affixed or mounted onto the front of the milling machine with the following components:

a. Front worktable made of stainless steel.
b. Control pendant.
c. Tooling holding vice mounted onto the machine made of stainless steel.
d. Tooling tray that supports 40 taper tool holders made of stainless steel.
e. Air pressure gun.
f. Stainless steel storage tray.
g. High-intensity lighting with limited heat output (example: LED lights) to illuminate, properly load and inspect workpiece.

3.2.12 Electrical control box. The electrical control box shall consist of a clipboard, work beacon, vise handle holder, storage pull down access door, tool tray, G- and M- code reference list, operators manual, assembly data, and remote jog handle.

3.2.12.1 Hardware features. The CNC system shall include, as a minimum, the following hardware:

a. Ethernet interface.
b. Wi-Fi connectivity between the controller and the local area network.
c. 32 Gigabyte memory or greater Allowing storage of large programs and media files, including .jpeg, .png, .pdf, .mov, and .mp4.
d. Control touch-screen monitor.
e. One or more 32-bit or 64-bit microprocessors with math coprocessor.
f. Universal Series Bus (USB) port.
g. IBM-style keyboard or alpha-numeric keyboard with dedicated keys for specific machine functions (see 3.2.12.3 through 3.2.12.6).
h. Pendant for machine remote control from the worktable area.
i. Work light internally mounted in the machine.

3.2.12.2. Data package. An external data package with maintenance and technical support shall include, as a minimum, the following hardware and software:

a. Three (3) year multi-user subscription and/or license to most current version of Autodesk – Fusion 360 with FeatureCAM Ultimate design and manufacturing software.
b. Three (3) year subscription and/or license to most current version of SOLIDWORKS Professional design and modeling software.
c. Personal computer (PC) with Windows 10 operating system. Multiple core 64-bit processor, 16GB RAM, 2GB OpenFL 2.0 complaint graphics card, 1920 x 1200 screen resolution, 16GB disk space.
d. All hardware and software shall be capable and compatible with machine controllers and post-processors.

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3.2.12.3 Control panel. Unless otherwise specified, the CNC system shall include, as a minimum, a control panel with the following manual controls and switches:

a. Emergency stop. Press to stop all axis motion, disable servos, stop the spindle and tool changer, and turn off the coolant pump.
b. Handle jog.
c. Cycle start. Starts a program. This button is used to start a program simulation in graphics mode.
d. System keyed lock.
e. Power on and off switches.
f. CNC system start/stop switches (cycle start/stop switches).
g. Spindle start/stop switches; spindle motor power and spindle speed meters.
h. Spindle speed override rotary switch.
i. Feed hold. Stops all axis motion during a program.
j. Feed rate override.
k. Jog and jog direction.
l. Setup mode. In the locked position, this key switch prevents alterations to programs, settings, parameters, and offsets.
m. Second home. Press to rapidly move all axes to coordinate as specified in the settings.
n. Auto door override. Press this button to operate or close the auto door (if equipped).
o. Memory lock. In the locked position, this key switch prevents alterations to programs, settings, parameters, and offsets.
p. USB. Plug in compatible USB devices into this port. Shall be equipped with a dust cover.

3.2.12.3.1 Mill keyboard.

a. Function keys:
    1. Reset – Clears alarms. Clears input text. Sets overrides to default values if Setting 88 is ON.
    2. Power up – Zero returns all axes and initializes the machine control.
    3. Recover – Enters tool changer recovery mode.
    4. Tool Offset Measure – Records tool length offsets during part setup.
    5. Next Tool – Selects the next tool from the tool changer.
    6. Tool Release – releases the tool from the spindle when in MDI, ZERO RETURN, or HANG JOG mode.
    7. Part Zero Set – records work coordinate offsets during part setup.
b. Cursor key:
    1. Home – Moves the cursor to the top-most item on the screen; in editing, this is the top left block of the program.
    2. Cursor Arrows – Moves one item, block, or field in the associated direction.
    3. Page up, page down – Used to change displays or move up/down one page when viewing a program.
    4. End – Moves the cursor to the bottom-most item on the screen. In editing, this is the last block of the program.
    5. Program – Selects the active program pane in most modes.

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6. Position – Selects the position display.
7. Offsets – Displays the Tool Offset and Work Offset tabbed menu.
8. Current Commands – Displays the menus for Devices, Timers, Macrons, Active Codes, Calculators, Advanced Tool Management, Tool Table and media.
9. Alarms – Displays the Alarm viewer and Message Screens.
10. Diagnostics – Displays tabs for Features, Compensations, Diagnostics, and Maintenance.
11. Settings – Displays and allows changing of user settings.
12. Help – Displays help information.

c. Display keys:
   1. Program – Sets the active program in most modes.
   2. Position – Selects the positions display.
   3. Offsets – Displays the Tool Offset and Work Offset tabbed menu.
   4. Current Commands – Displays menus for Devices, Timers Macrons, Active Codes, Calculators, Advanced Tool Management (ATM), Tool Table, and Media.
   5. Alarms – Displays the Alarm viewer and Message Center.
   6. Diagnostics – Displays tabs for Features, Compensation, Diagnostics, and Maintenance.
   7. Settings – Displays and allows changing of user settings.
   8. Help – Displays help information.

d. Mode keys:
   1. Edit – Lets operator edit programs in the editor.
   2. Insert – Enters text from the input line or the clipboard into the program at the cursor position.
   3. Alter- Replaces the highlighted command or text from the input line or the clipboard.
   4. Delete – deletes the item that the cursor is on, or deletes a selected program block.
   5. Undo – Undoes up to the last 40 edit changes and deselects a highlighted block.

e. Jog keys:
   1. Chip auger forward – Starts the chip removal system in the forward direction (out of the machine).
   2. Chip auger stop – Stops the chip removal system.
   3. Chip auger reverse – Starts the chip removal system in the “reverse” direction.
   4. Axis jog keys – Jog axes manually.
   5. Jog lock – Works with the axis jog keys.
   6. Coolant up – Moves the optional programmable coolant nozzle up.
   7. Coolant down – Moves the optional programmable coolant nozzle down.

f. Override keys: Allows the operator to change a parameter, without the need for altering the part program.
   1. 10 percent feed rate – Decreases the current feed rate by 10 percent.
   2. 100 percent feed rate – Sets an overridden feed rate back to the programmed feed rate.
   3. Handle control feed rate – Let operator use jog handle to adjust the feed rate in 1 percent increments.

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4. 10 percent spindle – Decreases the current spindle speed by 10 percent.
5. 100 percent spindle – Sets the overridden spindle speed back to the programmed speed.
6. 10 percent spindle – increases the current spindle speed by 10 percent.
7. Handle spindles – Lets you use the jog handle to adjust the spindle speed in 1 percent increments.
8. Forward – Starts the spindle in clockwise direction.
9. Stop – Stops the spindle.
10. Reverse – Starts the spindle in the counter clockwise direction.

3.2.12.3.2 Control display. The basic control layout shall include the following:

a. Mode, Network and Time Status Bar.
b. Program Display.
c. Active Codes.
d. Active Tools.
e. Coolant.
f. Alarm Status.
g. System Status Bar.
h. Position Display / Axis Load.
i. Input Bar.
j. Icon Bar.
k. Spindle Status.

3.2.12.3.3 Standard calculator. The calculator tab shall include calculators for basic mathematical functions, milling and tapping. For example, addition, subtraction, multiplication and division as well as square root and percentage. The calculator lets you easily transfer operations and results to the input line so that you can put them into programs. You can also transfer results into the Milling and Tapping calculators.

3.2.12.4 Operating modes. The CNC system shall include, as a minimum, the following operating modes:

a. Manual operation mode.
b. Manual data input mode.
c. Setup mode.
d. Learn mode.
e. Transmit and receive mode.
f. Conversational mode.
g. Auto mode.
h. Machine diagnostic/maintenance/monitoring modes.
i. Background editing mode/split screen program preview.

3.2.12.5 Axis and tool management. The CNC system shall include, as a minimum, the following axis and tool management features:

a. Tool length compensation.

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b. Tool life management.
c. Cutter diameter compensation.
d. Tool breakage detection.
e. Scale feedback.
f. Backlash compensation.
g. Tool offsets.
h. Number of tool offsets.
i. Tool number.
j. Type of tool.
k. Maximum load (highest load, in percent, that has been put on the tool).
l. The remaining percentage of the tool life.

3.2.12.6 Software features. The CNC system shall include, as a minimum, the following software programming and control features:

a. Simultaneous control of X-, Y-, and Z-axes and up to two optional rotary axes.
b. Linear, circular, parabolic, and helical interpolation.
c. Inch/metric switchable programming.
d. Decimal point programming.
e. Workpiece graphic display rotation.
f. Tool path graphic display.
g. Reference zero.
h. Standard G-codes, M-codes, and T-codes.
i. Standard suite of preprogrammed cycle software.
j. Dimensional absolute/incremental programming.
k. Cartesian/polar coordinate programming.
l. Parametric programming.
m. High-Speed Machining processor option.
n. Renishaw probe.

3.2.12.7 Variable data block format. The preparatory function G-codes, miscellaneous function M-codes, and tool function T-codes shall be in accordance with SAE RS-274.

3.2.12.8 Battery backup. The CNC system shall have a volatile memory battery backup (dual battery configuration) for an uninterrupted power supply (UPS) with a storage capacity of not less than 72 hours.

3.2.12.9 Lubrication system. The machining center shall be equipped with an automatic lubricating system to direct fluid flow from the reservoir to all machine components requiring continuous lubrication. Machine parts requiring periodic manual lubrication shall be readily accessible through doors or removable panels.

3.2.13 Coolant system. Cutting tools shall be flood cooled, and the machine shall be equipped with a system to control the rate of clean coolant flow from the reservoir and to direct coolant flow to nozzles mounted on the spindle head. The coolant level shall be displayed on the main screen and shall indicate if it is on or off.

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3.2.14 Electrical system. The electrical system shall conform to the ANSI/National Fire Protection Association (ANSI/NFPA) 79. See Table I for additional details.

3.2.15 Electric motors. The spindle drive motor horsepower shall correspond to a one-hour duty cycle rating. All motors shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1. The motors shall have open drip proof enclosures. Motion and position control motors and controls shall conform to NEMA ICS 1, NEMA ICS 3, and NEMA MG-7, unless otherwise agreed to by the Government (example: Air Force Metals Technology Office).

3.3 Performance.

TABLE I. Minimum Machine Parameters and Performance Requirements.

Parameters / Performance		***Type I***	***Type II***	***Type III***	***Type IV***	***Type V***
Spindle	Rating, HP	7-8	7-8	29-31	23-25	29-31
	Speed, rpm	5900-6000	5900-6000	7900-8200	19500-21500	7900-8200
	Cooling	Air Cooled	Air Cooled	Liquid Coolant	Liquid Coolant	Liquid Coolant
	Taper	CT or BT 40	CT or BT 40	CT or BT 40	HSK-A63	CT or BT 40
	Torque	32-34 ft-lbs @1,200 rpm	32-34 ft-lbs @1,200 rpm	89-95 ft-lbf @ 2,000 rpm	10-11 ft-lbf @ 12,000 rpm	88-92 ft-lbs @ 2,000 rpm
	Nose to Table or Platter (max), in	13-15	19-22	29-30	Varies with spindle orientation	23-25
	Nose to Table or Platter (min), in	3-5	3.9-4.1	4.125-4.25	Varies with spindle orientation	3.75-4.25

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TABLE I. Minimum Machine Parameters and Performance Requirements Cont.

Parameters / Performance		***Type I***	***Type II***	***Type III***	***Type IV***	***Type V***
B-Axis - Tilt	Travel, degrees	N/A	N/A	N/A	120 ° to 120°-	201° to 35-°
	Max Speed, degree/second	N/A	N/A	N/A	110-115	49-51
	Brake Torque, ft-lbf	N/A	N/A	N/A	400-410	1950-2100
	Max Torque, ft-lbf	N/A	N/A	N/A	290-300	2200-2300
C Axis - Rotation	Travel, degree	N/A	N/A	N/A	245° to 245-°	360
	Max Speed, degree/second	N/A	N/A	N/A	208	49-51
	Max Part Swing	N/A	N/A	N/A	N/A	26-29 in
	Brake Torque, ft-lbf	N/A	N/A	N/A	N/A	1800-1900
	Max Torque, ft-lbf	N/A	N/A	N/A	N/A	850-980

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TABLE I. Minimum Machine Parameters and Performance Requirements Cont.

Parameters / Performance		***Type I***	***Type II***	***Type III***	***Type IV***	***Type V***
Table or Platter	Length, in	35-36.5	57-58	51-54	62-66	N/A
	Width, in	28-29	10-10.75	18-19	35-37	N/A
	Diameter, in	N/A	N/A	N/A	N/A	19.5-20
	T-Slot Width, in	.62-.63	.62-.63	.63-.68	.63-.68	.63-.68
	T-Slot Center Distance, in	4.3-4.37	3.9-4.1	3.125-3.25	4.87-5	2.25-2.55
	Max Weight on Table, evenly distributed, lbf	490-510	980-1010	3200-3800	3800-4200	600-680 lbs
	Number of Standard T-Slots	3	3	5	7	7
Feed rates	Max Cutting, inch per minute	500	395-400	630-670	490-510	640-660
Tool Changer	Capacity (min)	10	10	20	50+1	30+1
	Max Tool Diameter	3.5	3.5	3.375-3.63	4.9-5.1	4.9-5.1
	Type	Carousel	Carousel	Carousel	Side Mounted Tool Changer	Side Mounted Tool Changer
Coolant	Capacity, gallon	40	20	52-58	90-100	50-60
Electrical System	Input AC Voltage (3 Phase) – Low	220	220	220	220	220
	Full Load Amps (3 Phase) – Low	25	25	70	70	70

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TABLE I. Minimum Machine Parameters and Performance Requirements Cont.

Parameters / Performance		***Type I***	***Type II***	***Type III***	***Type IV***	***Type V***
Axis Motors	Max Thrust X, lbf	1975-2050	1950-2100	2500-2600	3200-3500	2700-2800
	Max Thrust Y, lbf	1975-2050	1950-2101	2500-2600	3200-3500	2700-2800
	Max Thrust Z, lbf	1975-2050	1950-2102	4175-4200	5500-5700	3350-3450
Axis		3	3	3	5	5
Dimensions - Shipping - Domestic	Pallet, in x in x in (max)	90 x 64 x 100	130 x 78 x 99	152 x 99 x 106	195 x 120 x 140	168 x93 x 110
	Max Weight, (lbf)	4050	5250	12000	25250	14250
Dimension - Shipping - International	Pallet, in x in x in (max)	90 x 64 x 100	130 x 78 x 99	131 x 92 x 101	195 x 120 x 140	169 x 92 x 100
	Max Weight, (lbf)	4050	5250	12000	25250	14250
Linear Displacement Accuracy (min)	X, Y and Z directions, (inches/40 inches)	0.0009	0.0009	0.0007	0.0005	0.0005
Spindle thermal growth (long term, more than 4 hours)	Without Spindle Chiller (inches) / With Spindle Chiller, if equipped (inches)	0.003 / 0.0015	0.003 / 0.0015	0.003 / 0.0015	0.003 / 0.0015	0.003 / 0.0015
Bidirectional repeatability	X, Y and Z (inches/40 inches)	± 0.0001	± 0.0001	± 0.0001	± 0.0001	± 0.0001
Volumetric performance	Diagonal displacement (inches/40 inches)	0.0005	0.0009	0.0007	0.0005	0.0005
Squareness Error	X-Y, Y-Z, and Y-Z axes, (arcseconds)	3	3	3	3	3

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TABLE I. Minimum Machine Parameters and Performance Requirements Cont.

Parameters / Performance		***Type I***	***Type II***	***Type III***	***Type IV***	***Type V***
Contouring tolerance (up to 80% maximum speed)	X-Y plane, 360° (inches)	0.0009	0.0009	0.0009	0.0009	0.0009
	Y-Z plane, 190° (inches)	0.0009	0.0009	0.0009	0.0009	0.0009
	X-Z plane, 190° (inches)	0.0009	0.0009	0.0009	0.0009	0.0009
Environmental conditions and requirements	Operating temperature, (° F)	41-122	41-122	41-122	41-122	41-122
	Storage temperature, (° F)	-4 - 158	-4 - 158	-4 - 158	-4 - 158	-4 - 158
	Ambient Humidity (% relative, non-condensing)	20-90	20-90	20-90	20-90	20-90
Compressed Air Requirements	Pressure (psi), min	80	80	80	80	80

3.3.1 Machine tooling and accessory equipment. The contractor shall deliver tooling in Table II in accordance with equipment type identified in Table I.

TABLE II. Minimum Tooling and Accessory Requirements.

Tool / Accessory	***Quantity***	***Type I***	***Type II***	***Type III***	***Type IV***	***Type V***
CAT40 ER16 Collet Chuck, 2-1/2" length	2	X	X	X	X (or HSK-63A)	X
CAT40 ER16 Collet Chuck, 4" length	1	X	X	X	X	X
ER16 Collet Set, 5-10 pieces, sizes ranging from .040" to .390"	1	X	X	X	X	X
ER16 Collet Wrench	1	X	X	X	X	X
CAT40 ER25 Collet Chuck, 4" length	2	X	X	X	X	X
ER25 Collet Set, 10 - 20 pieces, sizes ranging from .080" to .625"	1	X	X	X	X	X

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TABLE II. Minimum Tooling and Accessory Requirements cont.

Tool / Accessory	***Quantity***	***Type I***	***Type II***	***Type III***	***Type IV***	***Type V***
ER25 Collet Wrench	1	X	X	X	X	X
CAT40 ER32 Collet Chuck, 2-1/2" length	4	X	X	X	X	X
CAT40 ER32 Collet Chuck, 4" length	2	X	X	X	X	X
ER32 Collet Set, 10-20 pieces, sizes ranging from .125" to .781"	1	X	X	X	X	X
ER32 Collet Wrench	1	X	X	X	X	X
CAT40 Shell Mill holder, 3/4" pilot hole, 2" length	2	X	X	X	X	X
CAT40 Shell Mill holder, 1" pilot hole, 2" length	2	X	X	X	X	X
Shell Mill Body, 3.0" diameter, 3/4" mount bore, Face Milling, 45° Cutting Edge, Positive Radial and Axial Rake, holds minimum of five screw clamping cutting inserts, .130" maximum depth of cut	2	X	X	X	X	X
HN25 Grade Cutting Insert	10	X	X	X	X	X
HK25 Grade Cutting Insert	10	X	X	X	X	X
Shell Mill Body, 3.0" diameter, 1" mount bore, Face Milling, Shoulder Milling, 90° Cutting Edge, holds minimum of five screw clamping cutting inserts, .300" maximum depth of cut	2	X	X	X	X	X
CAT40 3/4" End mill holder, 1-3/4" length	3	X	X	X	X	X
CAT40 1" End mill holder, 1-3/4" length	2	X	X	X	X	X
CAT40 Pull Stud set, 5-10 pieces	2	X	X	X	X	X
6" (160mm) Rotary Table, minimum accuracy of ± 16 arc-sec, minimum resolution of 0.0015 °, capable of plug-in connection to machine	1	X				X

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TABLE II. Minimum Tooling and Accessory Requirements cont.

Tool / Accessory	***Quantity***	***Type I***	***Type II***	***Type III***	***Type IV***	***Type V***
8" (210mm) Rotary Table, minimum accuracy of ± 16 arc-sec, minimum resolution of 0.0015 °, capable of plug-in connection to machine	1		X			X
Manual, Fixed Jaw Vice, 6" width, capable of mounting to machine T-slots, minimum of two T-slot alignment keys, 3/4 (16mm) wrench size	2	X	X	X	X	X
Spindle Taper Cleaner, 40-Taper	1	X	X	X	X	X

3.4 Machine axes. The machining center linear and rotary axes shall be identified in accordance with EIA RS-267.

3.5 Machine dimensions and weight. Machine dimensions and weight shall be in accordance with Table I.

3.6 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B 11.8. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910. If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall take precedence.

3.7 System of units. The machine shall be capable of supporting U.S. Customary System of Units (US) and the International System of Units (SI) to graduate measuring and indicating devices. Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. Measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.8 Nameplate. A nameplate shall be securely attached to the machine. At a minimum, the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Serial number.
d. Machine model designation.
e. Power input (volts, total amperes, phase, and frequency).
f. Short-circuit/over-current rating.
g. Contract number or order number.
h. National stock number.
i. Date of manufacture.

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3.9 Lubrication plate or chart. A lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

3.10 Equipment training. A complete overview of the machine controller, graphics display, and machine function and operation shall be provided at the installation location. Training shall be coordinated with the training point of contact (POC) at the installation location no later than seven calendar days after setup/commissioning at the installation location. The training POC will be provided by the government at the time of order. Training shall follow installation of the equipment and last no more than three days for a maximum of five technicians. Training shall provide the necessary skills and knowledge to operate the machine safely, execute manual data input functions, manual operation of machinery, load tooling into tool changer, conduct tooling offsets, and importing manufacturing files. Electronic versions of training materials shall be accessible to training participants for a minimum of three years after completion of the training.

3.11 Warranty. A minimum one year warranty shall be provided to protect against defects, breaks, or component failures during normal operations. The warranty shall include the replacement and installation of components/assemblies at the installation location by an OEM-certified technician. The warranty does not cover consumable items such as tooling identified in 3.3.1. For equipment located in austere operating environments, an OEM-certified technician may oversee the repair by a government representative; however, replacement components will still be provided as part of the warranty.

3.12 Support. Technical support for the equipment and software shall be provided for a period of three years after the date of purchase. Technical support shall be accessible, at a minimum, via phone and email.

4 REGULATORY REQUIREMENTS.

4.1 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials or bio-based content to the maximum extent practicable, in accordance with Federal Acquisition Regulation, (FAR) Part 23.403.

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user’s health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5 PROUDUCT CONFORMANCE.

5.1 Product conformance. The products shall meet the salient characteristics of this CID; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same machine offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

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5.2 Inspection. The machining center shall be inspected for compliance with all requirements specified in this CID.

5.3 Testing. The machining center shall be tested in accordance with sections 5.3.1 through 5.3.4. Testing shall demonstrate compliance with requirements herein and shall be verified by the cognizant Government Engineering Support Authority (Air Force Metals Technology Office / Metal Fabrication and Technologies Office).

5.3.1 Test setup, and conditions. The manufacturer shall be responsible for supplying all installation support, tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide all manpower required to support testing and all test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration. The machine shall be tested in accordance with environmental conditions specified in American Society of Mechanical Engineers (ASME) B5.54.

5.3.2 Operational test. The machining center and the CNC system shall be operated in accordance with the manufacturer's standard operational test procedures. The Government must approve any testing plans/procedures prior to testing occurring; the contractor shall be responsible for all related testing, test planning and the production of a final, comprehensive test reports. Proper operation of all controls and features shall be demonstrated and verified during this trial period by the Government (example: Air Force Metals Technology Office / Metal Fabrication and Technologies Office, etc.).

5.3.3 Performance tests. Performance tests shall be conducted in accordance with appendix B of ASME B5.54, or equivalent (Government concurrence shall be required.). The contractor shall work with the Government to ensure testing occurs per this specification and ASME B5.54.

5.3.4 Maximum horsepower cutting test. To demonstrate machine rigidity, performance, and capability, the machining center shall be tested in accordance with ASME B5.54, paragraph 9.5.8 (or ISO 10791-7:2014(E)), which specifies full-torque test cuts on a 4340 steel (300-350 BHN) workpiece with either a face or end mill. No evidence of chatter, stalling, or permanent distortion of machine components shall be permitted during or subsequent to the test.

5.3.5 Optional tests. Optional tests shall be conducted, by the contractor, as specified by the Government.

5.3.6 Acceptance. Preliminary and final acceptance tests shall be conducted at the manufacturer's site. Failure of the machining center to meet performance requirements shown in Table I, and in section 3 herein, for the machine type specified shall result in rejection.

5.3.7 Optional tests. Optional tests shall be conducted, by the contractor, as specified by the Government.

6. PACKAGING. Preservation, packing, and marking shall be as specified by the contract or order.

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7. NOTES.

7.1 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

A-A-60025-X

Type: A – Type I
B – Type II
C – Type III
D – Type IV
E – Type V

7.2 Sources of documents.

7.2.1 American National Standards Institute (ANSI). Copies of ANSI documents may be obtained at https://www.ansi.org/.

7.2.2 The American Society of Mechanical Engineers (ASME). Copies of ASME documents may be obtained at https://www.asme.org/.

7.2.3 SAE International. Copies of SAE documents may be obtained at https://www.sae.org/.

7.2.4 Federal Acquisition Regulation (FAR). Copies of FARS may be obtained at https://www.acquisition.gov/browse/index/far/.

7.2.5 National Electrical Manufacturers Association (NEMA). Copies of NEMA standards may be obtained at https://www.ansi.org/.

7.2.6 National Fire Protection Association (NFPA). NFPA documents may be obtained at https://www.nfpa.org/.

7.2.7 American National Standards Institute (ANSI). Copies of ANSI documents may be obtained at https://www.ansi.org/.

7.2.8 Occupational Safety and Health Administration (OSHA). Copies of OSHA documents may be obtained at https://www.osha.gov/.

7.3 NSNs

PIN	NSN
AA60025-A	3417-01-698-8845
AA60025-B	3417-01-698-8846
AA60025-C	3417-01-698-8847
AA60025-D	3417-01-698-8848
AA60025-E	3417-01-698-8850

7.4 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type of machining center (see 2).
c. Point of contact for training coordination at installation location (name, address, email, and telephone numbers).

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7.5 Key Words.
Milling
Cutting
Programmable
Multi-Axis

MILITARY INTERESTS:

Custodians:
Air force – 184

Reviewers:
GSA FAS

Preparing Activity:
Air Force – 184

Agent:
AF – 110

(Project 3417-2021-001)

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